README — bam_coverage.py

Purpose

bam_coverage.py scans a directory of STAR-aligned BAM files, computes total mapped reads per sample using samtools idxstats, and writes a simple TSV summary. It's a quick QC to flag low-coverage libraries before downstream analyses (e.g., strandedness inference, counting).

Requirements

• Python 3.7+
• samtools (in PATH)
• Readable BAM files with associated .bai index files (if missing, run samtools index sample.bam)

Inputs

• BAM directory (bam_dir in the script)
Default: /home/chu25/dsmz/results/star
• Minimum reads threshold (min_reads)
Default: 1,000,000 (mapped reads)

Outputs

• Coverage summary TSV (output_path)
Default: /home/chu25/dsmz/results/strandness/bam_coverage.tsv
TSV columns:
– sample — sample name derived from BAM filename (prefix before first dot)
– mapped_reads — total mapped reads (sum of column 3 from samtools idxstats)
– status — OK if mapped_reads ≥ min_reads, else LOW_COVERAGE (FAILED or ERROR if a run or parsing issue occurred)

How it Works

For each *.bam in bam_dir:
1. Runs samtools idxstats sample.bam
2. Sums the third column (mapped reads) across all references
3. Compares against min_reads
4. Writes a row to the output TSV:
samplemapped_readsstatus

Usage

1) Edit configuration (optional)
bam_dir = "/home/chu25/dsmz/results/star"
output_path = "/home/chu25/dsmz/results/strandness/bam_coverage.tsv"
min_reads = 1_000_000

2) Run
python bam_coverage.py
chmod +x bam_coverage.py && ./bam_coverage.py

Example Output

sample mapped_reads status
NG-13640_BC3_lib236396_5774_5 14234567 OK
NG-29643_BT_474_lib581288_7997_4 823410 LOW_COVERAGE

Interpreting Results

• OK — sample meets the minimum mapped reads threshold.
• LOW_COVERAGE — consider re-sequencing, dropping the sample, or interpreting results cautiously.
• FAILED/ERROR — check stderr/logs; likely samtools error or corrupt/locked BAM.

Troubleshooting

• “No BAM files found” — Verify bam_dir and filename pattern (*.bam).
• “idxstats failed” — Ensure BAMs are readable and indexed (.bai present). Reindex with samtools index.
• Mapped reads look too small — Confirm alignment pipeline and that BAMs are coordinate-sorted.

Notes

samtools idxstats counts mapped reads per reference sequence; the script sums these to get the per-sample total.
This script does not modify BAMs. It only reports coverage.
For batch/HPC use, wrap in a shell script or submit via your scheduler; the TSV serves as an artifact for downstream pipelines.